

Finite Quantum Dilogarithms: Normalization Exceptions, Infinitesimal Rigidity, and Effective Degree Bounds

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Abstract

We study the raw finite quantum-dilogarithm equations introduced by Radchenko–Wheeler. We classify their small-order normalization exceptions, compute the complete saturated solution schemes for the cyclic three-point metric and the anisotropic four-point metric, and give exact certificates for their reducedness. Using the known finiteness theorem, a Fourier–Weil eigenspace reduction gives explicit algebraic-degree bounds, with a sharper bound in prime order. We prove finite étaleness of the full raw solution scheme by extending the normalized Leavitt construction over the dual numbers and reconstructing its coordinates as monoidal invariants. The proof includes a corrected completeness coefficient, an explicit flatness argument, and normalized reconstruction with associators and tensorators. An elementary calculation also recovers a known obstruction for higher-rank elementary abelian two-groups.

Status and provenance. This is a research draft. The exact small-order certificates and degree bounds do not depend on the general finite-étale proof. The present revision discharges the dual-number realization hypothesis of the initial release; Theorem 3.1 concerns reducedness of the actual defining scheme. The argument is internally audited, not independently refereed or formally verified. The original working note was dated September 21, 2026. We compare the draft with Radchenko–Wheeler’s September 29 version [2], which restricts Theorem 5 to $N > 4$ and explicitly records small-order exceptions in Remark 3. The finite pentagon, algebraicity, and finiteness results are due to Radchenko–Wheeler, not to this note. The recent all-rank twisted-convolution theorem and modular-cocycle dictionary of Appleby–Flammia–Kopp [3] are likewise cited background. No priority claim is made for observing the small-order exceptions.

The companion functional paper [4] treats a different question: relations between quantum polylogarithms at fixed irrational parameter. Neither paper proves a numerical specialization theorem, Stark reciprocity, or the existence of the required unitary Galois conjugates.

1 The raw solution scheme

Let G be a finite abelian group of order $N > 2$, with an even quadratic function $q : G \rightarrow \mathbb{C}^\times$, and put

$$\chi(x, y) = \frac{q(x + y)}{q(x)q(y)}.$$

Assume χ is a nondegenerate bicharacter. Write $s = \sqrt{N} > 0$ and

$$\hat{f}(u) = \frac{1}{s} \sum_{v \in G} f(v) \chi(-u, v).$$

All values of q are roots of unity. Choose a number field $K \subset \mathbb{C}$ containing these values and s .

The equations used throughout are

$$\chi(x, y)E(x)E(y) = C + \frac{1}{s} \sum_{t \in G} q(t)E(x-t)E(t)E(y-t), \quad C \neq 0. \quad (1)$$

This is the raw equation of [1, Definition 4, Eq. (29)], with its bicharacter retained. Our equation, rather than any version-dependent equation number, fixes the scheme studied below. For $N \leq 4$ we do not impose the extra zero-coordinate normalization used to extend the definition in [2, Remark 3]; thus our raw scheme includes the exceptional points. Set

$$P_{xy} = s(\chi(x, y)X_x X_y - C) - \sum_t q(t)X_{x-t}X_t X_{y-t},$$

$$X_{G,q} = \text{Spec } K[X_g : g \in G, C, C^{-1}] / (P_{xy} : x, y \in G).$$

Equivalently one can introduce T and add $TC - 1$.

The known finiteness of complex solutions [2, Section 5] does not imply reducedness of this defining ideal. We keep the actual localized ideal, rather than replacing it by its radical, when discussing tangent spaces and finite étaleness.

2 Small-order normalization exceptions

The normalization formulas in [2, Theorem 5], valid for $N > 4$, are

$$e^2 = se + 1, \quad e = E(0), \quad (2)$$

$$E(u)E(-u) = q(u)^{-1} \quad (u \neq 0), \quad (3)$$

$$\hat{E}(u) = \lambda q(u)E(u), \quad C = -\lambda e, \quad \lambda^3 = \Gamma := \frac{1}{s} \sum_x q(x)^{-1}. \quad (4)$$

At the two smaller orders under consideration, the raw equation has the exceptions described below; compare [2, Remark 3].

2.1 An exact four-point counterexample

Let $G = (\mathbb{Z}/2\mathbb{Z})^2$ and

$$q(0) = 1, \quad q(x) = -1 \quad (x \neq 0), \quad E = q, \quad C = -1.$$

In the order $0, (1, 0), (0, 1), (1, 1)$ the bicharacter matrix is

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Thus the metric is nondegenerate. Its quadratic function has autocorrelation

$$\frac{1}{2} \sum_t q(x+t)q(y+t) = 2\delta_{x,y}.$$

Since $q(t)^2 = 1$, the cubic sum in (1) is exactly this quantity, while the left side is $q(x+y) = 2\delta_{x,y} - 1$. Consequently all sixteen ordered equations hold, with nonzero $C = -1$. But $e = 1$ gives $e^2 - 2e - 1 = -2$, and for nonzero u the asserted reflection identity reads $1 = -1$.

The first version's proof used nonconstancy of $q(x)E(x)$ on $G \setminus \{0\}$ for $N > 2$; here it is identically one. The current version avoids this step at these small orders by its $N > 4$ hypothesis and separate normalization convention [2, Theorem 5 and Remark 3].

This counterexample is algebraic. It is not a counterexample to the paper's algebraicity claim for Stark special values. Those values satisfy the normalizations separately through Theorem 2 and Proposition 10 of [1].

2.2 The order-three exception

Let $G = \mathbb{Z}/3\mathbb{Z}$, $r = (-1 + i\sqrt{3})/2$, and $q(1) = q(2) = r$. Then

$$E(0) = \frac{\sqrt{3} - i}{2}, \quad E(1) = E(2) = -\frac{\sqrt{3} + i}{2}, \quad C = r^2 \quad (5)$$

satisfies all nine equations (1). Again $e^2 - \sqrt{3}e - 1 = -2$. Complex conjugation gives the other metric. These statements follow by direct substitution; the script checks them in an exact number field, with no numerical tolerances.

2.3 Why there are no other normalization exceptions

Lemma 2.1. *For a complex solution of (1), all $E(u)$ and $\widehat{E}(u)$ are nonzero. If $q(u)E(u)$ is nonconstant on $G \setminus \{0\}$, then (2)–(4) hold. This implication also holds over the dual numbers in a neighbourhood of such a solution.*

Proof. We indicate explicitly the local-algebra issue. The Fourier transform of (1), with $p_x(z) = E(x-z)E(z)q(z)$, is

$$E(x)\widehat{E}(w-x) = C s \delta_0(w) + \widehat{E}(w)\widehat{p}_x(w). \quad (6)$$

The support argument and Fourier inversion in [1, proof of Theorem 5, (31)–(33)] give nonvanishing, and give

$$E(u)E(-u)q(u) = s e \delta_0(u) - C/\widehat{E}(0), \quad \widehat{E}(u) = \lambda q(u)E(u) \quad (u \neq 0).$$

The next convolution identity in that proof gives constants c, k such that $qE - c\widehat{E} = k\mathbf{1}$. For distinct nonzero u, v with $q(u)E(u) \neq q(v)E(v)$, subtraction yields $c\lambda = 1$, then $k = 0$. This extends the Fourier relation to zero. The algebra in the cited proof then yields

$$e^2 = (e^2 - se)(se + 1),$$

whose difference from the left side is $se(e^2 - se - 1)$. Since $se \neq 0$, this gives (2); reflection and $C = -\widehat{E}(0)$ follow, and the Fourier–Weil cubic gives $\lambda^3 = \Gamma$.

For the local assertion, let $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$ and reduce to the given solution. Every $E(u)$, $\widehat{E}(u)$ and C is now a unit. The chosen nonzero difference $q(u)E(u) - q(v)E(v)$ is also a unit. The Fourier and convolution calculations just described are algebraic identities and use only these invertible quantities and nonzero scalar constants. Thus the same argument works in A ; it does not infer reducedness from an identity that holds only at field-valued points. $\square$

Suppose now that $q(u)E(u) = B$ for all $u \neq 0$. Then $B \neq 0$ and

$$E = Bq^{-1} + (e - B)\delta_0.$$

The Gaussian Fourier identity $\widehat{q^{-1}} = \Gamma q$ shows that, for $u \neq 0$,

$$\widehat{E}(u) = B\Gamma q(u) + (e - B)/s.$$

But the partial Fourier relation obtained before the problematic sentence in [1] says $\widehat{E}(u) = \lambda B$ on this punctured group. Since $B\Gamma \neq 0$, the value $q(u)$ must be a constant r for $u \neq 0$. Orthogonality of any nontrivial row of χ gives

$$0 = 1 + r^{-2} + (N - 2)r^{-1}, \quad r + r^{-1} = -(N - 2).$$

As $|r| = 1$, this forces $N \leq 4$. For $N = 3$ it gives $r^2 + r + 1 = 0$. For $N = 4$ it gives $r = -1$, and the group must be $(\mathbb{Z}/2\mathbb{Z})^2$: a cyclic group of order four would have $q(2g) = q(g)^4 = 1$, a contradiction. The exact calculations in Section 5 show that the only constant-puncture solutions in these metrics are the two types already listed. We have proved the following corrected alternative.

Proposition 2.2. *Every complex solution of (1), with $N > 2$, either satisfies (2)–(4), or is the order-three solution (5) (or its conjugate), or the displayed four-point solution. In particular the normalization in Theorem 5 of [1] is valid without exceptions when $N \geq 5$, as in [2, Theorem 5].*

3 Infinitesimal rigidity and finite étaleness

Theorem 3.1 (Finite étaleness of the raw scheme). *For every metric group (G, q) of order $N > 2$, the scheme $X_{G,q}$ is finite étale over K (possibly empty). In particular, its actual localized defining ideal is radical.*

The proof uses the Leavitt construction and reconstruction of [2, Sections 5.4–5.5 and Appendix B], but requires their identities over a nonreduced base. We establish this extension below, including a correction to the displayed completeness calculation. We then prove flatness and invariance of the normalized reconstructed coordinates under strong monoidal equivalences. These arguments discharge the realization hypothesis in the initial release of this note.

3.1 Normalization of an infinitesimal solution

Let $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$, and let

$$E_A(g) = E(g) + \varepsilon h(g), \quad C_A = C + \varepsilon c_1$$

be a solution reducing to a nonexceptional complex point. Here *nonexceptional* means that $q(g)E(g)$ is nonconstant on $G \setminus \{0\}$. Lemma 2.1 gives the normalized equations in A . Writing $e_A = e + \varepsilon e_1$ and $\lambda_A = \lambda + \varepsilon \lambda_1$, their scalar parts give

$$(2e - s)e_1 = 0, \quad 3\lambda^2\lambda_1 = 0.$$

Since $(2e - s)^2 = N + 4$ and $\lambda \neq 0$, it follows that

$$e_A = e, \quad \lambda_A = \lambda, \quad C_A = -\lambda e = C. \tag{7}$$

Thus $d = -s/e$ and a chosen square root ν of d are constant nonzero scalars, and

$$d^2 = N(d+1), \quad se = -N/d, \quad \nu^2 = d. \quad (8)$$

The remaining tangent equations include

$$E(-g)h(g) + E(g)h(-g) = 0 \quad (g \neq 0), \quad (9)$$

$$\widehat{h}(g) = \lambda q(g)h(g), \quad (10)$$

$$\begin{aligned} \chi(x, y)(h(x)E(y) + E(x)h(y)) &= \frac{1}{s} \sum_t q(t)(h(x-t)E(t)E(y-t) \\ &\quad + E(x-t)h(t)E(y-t) + E(x-t)E(t)h(y-t)). \end{aligned} \quad (11)$$

We will show that these infinitesimal motions vanish by constructing a flat deformation whose normalized categorical coordinates are $E_A(g)$.

3.2 Identities over the universal normalized coefficient ring

Fix e, λ, d, ν as above. Let

$$R = \mathbb{C}[Z_g : g \in G]/J,$$

where J is generated by $Z_0 - e$, the Fourier relations $\widehat{Z}(g) - \lambda q(g)Z_g$, the reflection relations $q(g)Z_g Z_{-g} - 1$ for $g \neq 0$, and the raw equations with $C = -\lambda e$. Write $E_R(g)$ for the class of Z_g . We do not take the radical of J . Every normalized dual-number solution in this sector factors through R . All denominators used below belong to $\mathbb{C}^\times$; also every $E_R(g)$ is a unit by reflection and $E_R(0) = e$.

The following finite Fourier identities hold in R :

$$\widehat{E}_R(g) = \lambda q(g)E_R(g), \quad q(g)E_R(g)E_R(-g) = 1 + se\delta_{g,0}. \quad (12)$$

The Fourier transform on R^G is invertible: character orthogonality is an identity in the constant coefficient field. In particular, every use of Fourier inversion below is valid before reducedness is known.

Lemma 3.2 (Correlation and dual pentagon). *In R , for every $a \in G$,*

$$K_R(a) := \sum_t q(t)E_R(-t)E_R(t+a) = N\delta_{a,0} - \frac{N}{d}. \quad (13)$$

The dual pentagon takes the form

$$q(x-y)E_R(x)E_R(y) = -\lambda^{-1}e + \frac{1}{s} \sum_t q(y-t)E_R(y-t)E_R(t)q(x-t)E_R(x-t). \quad (14)$$

Proof. Taking the normalized Fourier transform in a gives

$$\widehat{K}_R(\xi) = s\lambda^{-1}E_R(-\xi)\widehat{E}_R(\xi) = s + Ne\delta_{\xi,0}.$$

Fourier inversion gives $K_R(a) = N\delta_{a,0} + se$, proving (13).

For the second identity, use the finite Fourier transformation in the proof of [2, Theorem 5(iv)]. More explicitly, set $\widetilde{E}_R = qE_R$ and $\widetilde{C} = \lambda^{-2}C = -\lambda^{-1}e$. Multiply the proposed dual equation by $\chi(y, w)$ and sum in y with measure $1/s$. Using $\widehat{\widetilde{E}_R}(u) = \lambda^{-1}E_R(-u)$, it becomes

$$\lambda^{-1}q(x)E_R(x)E_R(w-x) = s\widetilde{C}\delta_{w,0} + \frac{\lambda^{-1}E_R(w)}{s} \sum_z E_R(x-z)E_R(z)\chi(z, w)q(x-z).$$

After $z \mapsto x - z$, multiplication by λ and the elementary bicharacter identities, this is the Fourier transform of the raw pentagon equation in the other coordinate. All transformations are invertible over R . No complex conjugation or assertion about the zero set of J is used. $\square$

Proposition 3.3 (Universal Leavitt realization). *In $L_{2N}(R) = R \otimes_{\mathbb{C}} L_{2N}$, the formulas of [2, Section 5.4] define R -linear automorphisms α_g , units U_g , and a unital endomorphism ρ_R . They satisfy*

$$\alpha_g \alpha_h = \alpha_{g+h}, \quad U_g U_h = U_{g+h}, \quad \alpha_g \rho_R = \rho_R, \quad \rho_R \alpha_g = \text{Ad}(U_g) \rho_R, \quad (15)$$

$$\rho_R^2(x) = \sum_g S_g \alpha_g(x) S'_g + \sum_g T_g \rho_R(x) T'_g \quad (x \in L_{2N}(R)), \quad (16)$$

$$\rho_R(dS'_0) S_0 = 1, \quad dS'_0 \rho_R(S_0) = 1. \quad (17)$$

The S_g, T_g and their primed partners give split fusion maps. All these identities commute with homomorphisms of coefficient rings.

Proof. Here the Leavitt generators obey

$$S'_g S_h = T'_g T_h = \delta_{g,h}, \quad S'_g T_h = T'_g S_h = 0, \quad P_S + P_T = 1,$$

where $P_S = \sum_g S_g S'_g$ and $P_T = \sum_g T_g T'_g$. For clarity we record the formulas, in our notation:

$$\begin{aligned} \alpha_g(S_h) &= S_{g+h}, & \alpha_g(S'_h) &= S'_{g+h}, \\ \alpha_g(T_h) &= \chi(g, h)^{-1} T_h, & \alpha_g(T'_h) &= \chi(g, h) T'_h, \\ U_g &= \sum_h \chi(g, h)^{-1} S_h S'_h + \sum_h T_{h-g} T'_h. \end{aligned}$$

The generator images of ρ_R , writing $E = E_R$ in this proof, are

$$\begin{aligned} \rho_R(S_g) &= U_g \left(d^{-1} \sum_h S_h + \nu^{-1} \sum_h q(h) T_h T_{-h} \right) U_{-g}, \\ \rho_R(S'_g) &= U_g \left(d^{-1} \sum_h S'_h + \nu^{-1} \sum_h q(h)^{-1} T'_{-h} T'_h \right) U_{-g}, \end{aligned} \quad (18)$$

$$\begin{aligned} \rho_R(T_g) &= \frac{\lambda}{\nu s} \sum_{h,k} \chi(h-g, k) S_h T'_k + \frac{\lambda^{-1}}{q(g)s} \sum_{h,k} \chi(g+h, k)^{-1} T_k S_h S'_h \\ &\quad + \frac{1}{s} \sum_{h,k} q(h) E(h+g) \chi(g, k)^{-1} T_{h+k} T_{-h} T'_k, \end{aligned} \quad (19)$$

$$\begin{aligned} \rho_R(T'_g) &= \frac{\lambda^{-1}}{\nu s} \sum_{h,k} \chi(h-g, k)^{-1} T_k S'_h + \frac{\lambda q(g)}{s} \sum_{h,k} \chi(g+h, k) S_h S'_h T'_k \\ &\quad + \frac{1}{s} \sum_{h,k} \frac{q(g+h)}{q(h)} E(-h-g) \chi(g, k) T_k T'_{-h} T'_{h+k}. \end{aligned} \quad (20)$$

Generator relations and completeness. The four primed-unprimed products in Appendix B.1 of [2] reduce using character orthogonality, (12), and (13). For example,

$$\rho_R(S'_a) \rho_R(S_b) = U_a \left(d^{-2} \sum_h \chi(a-b, h) + \frac{N}{d} \delta_{a,b} \right) U_{-b} = \delta_{a,b},$$

and the mixed products vanish because their scalar factor is $s/d + e = 0$. In $\rho_R(T'_a)\rho_R(T_b)$ the $T_k T'_k$ coefficient is $\chi(a - b, k)(1/d + K_R(b - a)/N) = \delta_{a,b}$; the $S_h S'_h$ coefficient is $\delta_{a,b}$ by orthogonality. Thus this product too is $\delta_{a,b}$.

There is a coefficient error in the displayed completeness calculation in [2, Appendix B.1, printed p. 30]: the coefficient of P_S is printed as $N(N + d)/d^2$. Direct multiplication of (18)–(20) instead gives

$$\frac{N}{d^2} + \frac{N}{d} = \frac{N(1 + d)}{d^2} = 1. \quad (21)$$

The two contributions are respectively from the S -sum and the T -sum. After the mixed monomials cancel, the full expression is

$$\begin{aligned} & \sum_g \rho_R(S_g)\rho_R(S'_g) + \sum_g \rho_R(T_g)\rho_R(T'_g) \\ &= \left(\frac{N}{d^2} + \frac{N}{d}\right) P_S + \sum_k T_k P_S T'_k \\ & \quad + \sum_{h,k,r} \frac{q(h)}{q(r)} \left(\frac{1}{d} + \frac{1}{N} K_R(h - r)\right) T_{h+k} T_{-h} T'_{-r} T'_{r+k} \\ &= P_S + \sum_k T_k P_S T'_k + \sum_k T_k P_T T'_k = 1. \end{aligned}$$

Consequently ρ_R is a well-defined unital endomorphism over R . The corrected calculation proves the required identity; it does not change the generator formulas or the construction in [2].

Covariance and the fusion identity. The α and U identities and (15) follow by multiplying the displayed formulas and reindexing finite sums. To make explicit why the longer fusion verification also applies over R , we record its reductions from [2, Appendix B.2]. First they give

$$\rho_R(U_g) = \sum_h S_{h-g} S'_h + \sum_h \frac{q(h+g)}{q(h)} T_{h+g} U_g T'_h.$$

Its mixed matrix coefficients vanish by the Fourier relation; its S -coefficients use (21), and its T -coefficients use (13). The subsequent calculations for $\rho_R^2(S_g)$, $\rho_R^2(U_g)$, and $\rho_R^2(x)S_g = S_g \alpha_g(x)$ use these identities, covariance, and the same Fourier and correlation relations.

For the remaining T -calculation put $\hat{T}_g = s^{-1} \sum_h \chi(g, h) T_h$. In the key identity

$$\rho_R(T'_0 \rho_R(T_0)) \hat{T}_0 = \rho_R(T'_0) \hat{T}_0 \rho_R(T_0),$$

the three coefficient comparisons are the following:

$$\begin{aligned} \lambda q(h) E(h) &= s^{-1} \sum_k E(k) \chi(h, -k), \\ \frac{\lambda}{N} \sum_h q(h)^2 \chi(h, -k) E(-h) E(k - h) &= \frac{\lambda^2}{s} - \frac{\lambda^{-1}}{dq(k)}, \\ q(x - y) E(x) E(y) &= -\lambda^{-1} e + s^{-1} \sum_h q(y - h) E(y - h) E(h) q(x - h) E(x - h). \end{aligned}$$

The first is Fourier, and the third is (14). For the second, Fourier inversion reduces its left side to

$$\frac{\lambda^{-1}}{Nq(k)} \sum_r E(r) E(-r) \chi(r, k).$$

Now use reflection, $s^{-1} \sum_r q(r)^{-1} \chi(r, k) = \Gamma q(k)$, $\Gamma = \lambda^3$, and $N/d^2 - 1 = -N/d$.

All steps in these reductions are identities of finite sums in R . For completeness, the cancellation at the end of the argument is also valid over a ring with zero divisors: if $\rho_R(S'_g)v = \rho_R(T'_g)v = 0$ for every g , then

$$v = \sum_g \rho_R(S_g) \rho_R(S'_g) v + \sum_g \rho_R(T_g) \rho_R(T'_g) v = 0.$$

Applied to the remaining difference, the preceding identities and covariance give $\rho_R^2(T_0) \hat{T}_g = \hat{T}_g \rho_R(T_0)$ for every g . The finite Fourier matrix is invertible over R , so this gives the same identity with T_g instead of $\hat{T}_g$. Together with $T_h = U_{-h} T_0$ and the already verified $\rho_R^2(U_{-h})$ identity, it gives $\rho_R^2(T_h) T_g = T_g \rho_R(T_h)$. These are precisely the remaining split fusion equations. Completeness proves (16) on unprimed generators. Equality on the primed generators follows from uniqueness of their left inverses subject to completeness, again without cancelling a nonunit.

Finally (17) follows at once from (18) and the Leavitt relations. Thus the fusion and (co-)evaluation maps exist over R . This proof is an identity calculation in R , not an appeal to validity at its complex points; applying any coefficient homomorphism preserves it. $\square$

3.3 Flatness over the dual numbers

Base change along $R \rightarrow A$ gives ρ_A and all the split maps. In particular,

$$\rho_A^2 \cong \bigoplus_{g \in G} \alpha_g \oplus \rho_A^{\oplus N}, \quad \alpha_g \rho_A = \rho_A, \quad \rho_A \alpha_g \cong \rho_A. \quad (22)$$

We do not infer flatness from the mere existence of algebraic formulas.

Lemma 3.4 (Flat split skeleton). *The additive tensor category generated by α_g and ρ_A has free finite-rank Hom modules with the same ranks as its complex reduction. A choice of the split skeleton and fusion bases lifting the special-fiber choices presents it as a first-order deformation of the fusion category constructed in [2, Section 5.4].*

Proof. The complex category has scalar endomorphisms of the basic simple labels α_g, ρ , and zero morphisms between distinct such labels, by the Hom calculations in [2, Section 5.4]. In $L_{2N}(A) = A \otimes_{\mathbb{C}} L_{2N}$ write an intertwiner as $z = z_0 + \varepsilon z_1$. Between distinct basic labels, reduction gives $z_0 = 0$; the coefficient of ε in the intertwining equation then makes z_1 an intertwiner between the distinct reduced labels, so $z_1 = 0$. For an endomorphism of a basic label, z_0 is scalar. Subtracting that scalar identity, which is itself an exact intertwiner over A , reduces to the same argument and makes z_1 scalar. Hence

$$\text{End}_A(\alpha_g) = A, \quad \text{End}_A(\rho_A) = A, \quad \text{Hom}_A(S, T) = 0 \quad (S, T \text{ distinct basic labels}).$$

The explicit decompositions (22) now reduce every tensor word to a direct sum of those labels. Hom modules are therefore matrices over A of the required sizes. Idempotent completion adds no new types: finite projective modules over the local ring A are free. The split maps identify the tensor bifunctor with its fixed fusion-rule model; the transported associator reduces to that of the complex category. The snake identities give rigidity. Thus this is a flat first-order tensor deformation, without claiming semisimplicity over the nonfield A . $\square$

3.4 Intrinsic reconstruction and monoidal gauge invariance

We next extend the normalization in [2, Section 5.5]. It is important to retain the associator: our endomorphism realization is strict, but the skeleton used for deformation theory need not be. Write $\mathfrak{a}_{X,Y,Z} : (X \otimes Y) \otimes Z \rightarrow X \otimes (Y \otimes Z)$. Identity morphisms and unit identifications are suppressed where unambiguous.

Lemma 3.5 (Normalized scalar reconstruction). *For the category of Lemma 3.4, and for any of its A -linearly strongly monoidally equivalent labelled models, the scalars*

$$\mathcal{I}_g 1_\rho = T'_0(T'_0 \otimes 1) \mathfrak{a}_{\rho,\rho,\rho}^{-1}(1 \otimes T_g) T_0 \quad (23)$$

are intrinsically defined by the normalization below. They are invariant under such equivalences and, in the Leavitt realization, satisfy

$$\mathcal{I}_g = E_A(g)/s. \quad (24)$$

Proof. Set $V_A = \text{Hom}_A(\rho, \rho \otimes \rho) = \bigoplus_h AT_h$ and $W_A = \text{Hom}_A(\mathbf{1}, \rho \otimes \rho) = AS_0$. Choose isomorphisms $\ell_g : \alpha_g \otimes \rho \rightarrow \rho$ and define

$$D_g(t) = (\ell_g \otimes 1) \mathfrak{a}_{\alpha_g, \rho, \rho}^{-1}(1 \otimes t) \ell_g^{-1}. \quad (25)$$

Rescaling ℓ_g by any unit leaves D_g unchanged. In the Leavitt model take $\ell_g = 1$; then $D_g(T_h) = \chi(g, h)^{-1} T_h$. Therefore

$$P_0 = \frac{1}{N} \sum_g D_g, \quad P_0(T_h) = \delta_{h,0} T_h, \quad \text{im } P_0 = AT_0.$$

This proves the rank-one assertion over A , rather than merely after reduction. Choose a *generator* T_0 of this free line, equivalently an element with nonzero reduction. A merely nonzero element such as εT_0 would not suffice. Any two generators differ by a unit.

Choose also $v_g : \rho \rightarrow \rho \otimes \alpha_g$ and set

$$B_g = (1 \otimes \ell_g) \mathfrak{a}_{\rho, \alpha_g, \rho}(v_g \otimes 1), \quad U_g = \eta_g^{-1} B_g, \quad (26)$$

where $\eta_g \in A^\times$ is the scalar by which B_g acts on W_A . It is a unit since B_g is invertible. The spaces containing ℓ_g and v_g are free of rank one, so any changes multiply B_g and η_g by the same unit. Thus U_g is canonical: it acts identically on W_A . In the Leavitt model this recovers the displayed U_g , since $U_g S_0 = S_0$. Now define $T_g = U_{-g} T_0$. These give a basis of V_A , determined up to one common unit. The dual basis $T'_g \in \text{Hom}_A(\rho \otimes \rho, \rho)$ is determined by $T'_g T_h = \delta_{g,h} 1_\rho$; the perfectness of this pairing follows from the split fusion decomposition.

Consequently the four factors in (23) change by u^{-1}, u^{-1}, u, u , so their product is unchanged. This cancellation would fail for unrelated basis changes $T_g \mapsto u_g T_g$: the factor would then be u_g/u_0 . The normalized U_g are what exclude that ambiguity.

Let F be an A -linear strong monoidal equivalence with tensorator $J_{X,Y} : F(X) \otimes F(Y) \rightarrow F(X \otimes Y)$, after identifying the simple labels in its source and target. Transport the maps by

$$\tilde{T}_g = J_{\rho,\rho}^{-1} F(T_g), \quad \tilde{T}'_g = F(T'_g) J_{\rho,\rho}.$$

Naturality and coherence transport (25) and its projector, so they transport the distinguished line. They also transport B_g and, using the unit constraint of F , its action on W_A . The uniquely normalized U_g therefore transport to the uniquely normalized operators in the target. The transported basis is consequently the target normalized basis up to its permitted common unit.

The associator–tensorator identity is

$$F(\mathbf{a}_{X,Y,Z})J_{X \otimes Y, Z}(J_{X,Y} \otimes 1) = J_{X,Y \otimes Z}(1 \otimes J_{Y,Z})\mathbf{a}_{F(X), F(Y), F(Z)}.$$

Using it in (23) gives

$$F(\mathcal{I}_g 1_\rho) = \tilde{T}'_0(\tilde{T}'_0 \otimes 1)\mathbf{a}_{F(\rho), F(\rho), F(\rho)}^{-1}(1 \otimes \tilde{T}_g)\tilde{T}_0.$$

Because F is A -linear, the scalar on the left is still $\mathcal{I}_g$. Thus the equality of reconstructed scalars holds in A , not only modulo ε . No strictness or preservation of a pivotal structure is required.

Finally, in the strict endomorphism model $1_\rho \otimes T_g = \rho_A(T_g)$. In $T'_0 \rho_A(T_g) T_0$, the first two sums of (19) vanish by the mixed Leavitt relations. The right T_0 forces $k = 0$ in the third sum, and the left T'_0 then forces $h = 0$. This leaves $E_A(g)T_0/s$. A further multiplication by T'_0 gives (24), as an identity in $L_{2N}(A)$. $\square$

Corollary 3.6 (Dual-number realization and reconstruction). *Every dual-number solution at a nonexceptional complex point admits the normalized Leavitt, split fusion, and duality realization over A . Its scalar reconstruction is compatible with this base change and with label-preserving A -linear strong monoidal equivalences reducing to the identity. Thus the normalized realization hypothesis of the initial release is satisfied.*

Proof. Combine (7), Proposition 3.3, and Lemmas 3.4 and 3.5. $\square$

3.5 Vanishing of tangent spaces and descent

Proof of Theorem 3.1. At a nonexceptional point, Lemma 3.4 gives a flat first-order deformation of the complex fusion category $\mathcal{C}$. For fusion categories in characteristic zero, the first-order associator deformation space is $H^3(\mathcal{C})$, and it vanishes by [5, Theorem 2.27 and Section 7.1]. On the fixed split skeleton there is therefore an A -linear strong monoidal equivalence, reducing to the identity, from our deformation to $A \otimes_{\mathbb{C}} \mathcal{C}$. The simple labels are fixed, so no relabelling by $\text{Aut}(G)$ occurs. Lemma 3.5 now gives

$$E_A(g)/s = \mathcal{I}_g(\mathcal{C}_A) = \mathcal{I}_g(A \otimes_{\mathbb{C}} \mathcal{C}) = E(g)/s.$$

Thus $h(g) = 0$ for every g , and $c_1 = 0$ by (7). Every nonexceptional tangent space is zero.

The only exceptional points are those of Proposition 2.2. At the order-three point a maximal Jacobian minor of the raw residuals in variables $(E(0), E(1), E(2), C)$ is

$$-\frac{27\sqrt{3}}{2} - \frac{81i}{2} \neq 0;$$

its conjugate handles the conjugate metric. At the four-point exception a maximal minor is 512. Section 5 and the exact verifier supply these certificates. Hence every closed complex point has zero Zariski tangent space.

At each such point the maximal ideal of its Noetherian local ring satisfies $\mathfrak{m}/\mathfrak{m}^2 = 0$, so Nakayama's lemma gives $\mathfrak{m} = 0$. A finite-type affine complex scheme with this property is zero-dimensional and reduced, hence finite over $\mathbb{C}$. Faithfully flat descent from $\mathbb{C}$ to K , together with characteristic-zero separability, gives finite étaleness over K . This concerns the original localized ideal, not an ideal replaced by its radical. $\square$

Corollary 3.7. *For every local Artin $\mathbb{C}$ -algebra B with residue field $\mathbb{C}$, reduction induces a bijection*

$$X_{G,q}(B) \longrightarrow X_{G,q}(\mathbb{C}).$$

In particular, the raw solution scheme has no hidden nilpotent motions of any order.

Proof. Apply formal étaleness to the nilpotent maximal ideal of B . $\square$

4 Effective degree bounds from Fourier–Weil eigenspaces

The results of this section use the established finiteness theorem [2, Section 5], not Theorem 3.1 or reducedness. Fix choices e, λ satisfying (2) and $\lambda^3 = \Gamma$, and write $K' = K(e, \lambda)$. Define the Fourier–Weil operator and its eigenspace dimension by

$$Wf(u) = q(u)^{-1} \widehat{f}(u), \quad m_\lambda = \dim_{K'} \ker(W - \lambda).$$

The identity $W^3 = \Gamma I$ is proved in [1, proof of Theorem 5]. In this fixed sector every normalized solution lies in

$$\ker(W - \lambda) \cap \{X_0 = e\}.$$

If the sector is nonempty, this is an affine space of dimension $m_\lambda - 1$; the functional $X \mapsto X_0$ is then nonzero on the eigenspace.

Proposition 4.1. *For a normalized finite quantum dilogarithm in a nonempty (e, λ) sector,*

$$[K'(E(g) : g \in G) : K'] \leq 3^{m_\lambda - 1}.$$

When $G = \mathbb{Z}/p\mathbb{Z}$ with p prime, the sharper bound is

$$[K'(E(g) : g \in G) : K'] \leq 2^{m_\lambda - 1}.$$

The same bounds hold for the number of geometric solutions in that sector.

Proof. In the affine eigenspace section the remaining defining equations have degree at most three. Its finite solution set has at most $3^{m_\lambda - 1}$ points by the affine Bezout bound. The field degree is the size of the orbit of the entire tuple under the absolute Galois group of K' , so it is bounded by that number. In prime order, [2, Section 5.1, prime-order uncertainty argument] shows that the larger variety cut out just by the linear Fourier equations and quadratic reflection identities is finite: at infinity, a vector and its Fourier transform would have equal skew support of size at most $(p - 1)/2$, contrary to the prime-order uncertainty principle. Consequently quadratic, rather than cubic, Bezout applies on the same affine space. $\square$

The dimensions m_λ are explicitly computable from a finite matrix. They also admit a uniform estimate.

Lemma 4.2. *Let $G[3] = \{g : 3g = 0\}$. Then*

$$m_\lambda \leq \left\lfloor \frac{N + 2\sqrt{|G[3]|}}{3} \right\rfloor.$$

Proof. The evenness of q implies $\chi(x, x) = q(x)^2$ and $q(nx) = q(x)^{n^2}$. Therefore

$$\mathrm{tr} W = \frac{1}{s} \sum_x q(x)^{-3}.$$

The radical of χ^3 is $G[3]$. For $z \in G[3]$, $q(z)^9 = q(z)^6 = 1$, so $q(z)^3 = 1$. The usual direct calculation of the squared absolute value of a Gaussian sum thus gives

$$\left| \sum_x q(x)^{-3} \right|^2 = N|G[3]|, \quad |\mathrm{tr} W| = \sqrt{|G[3]|}.$$

For completeness, expand the square using $x = y + z$, sum first over y , and use character orthogonality; only $z \in G[3]$ survive.

Choose $\lambda_0^3 = \Gamma$ and $\omega = e^{2\pi i/3}$. Since $W^3 = \Gamma I$ and $|\Gamma| = 1$, the eigenvalues are $\lambda_0 \omega^j$ with multiplicities m_j . If $T = \text{tr}(W)/\lambda_0$, Fourier inversion on three points gives

$$m_j = \frac{N + 2 \text{Re}(\omega^{-j} T)}{3}.$$

The asserted bound follows. $\square$

Corollary 4.3. *For $p \geq 5$ prime, a fixed metric q on $\mathbb{Z}/p\mathbb{Z}$, and $K \supset \mathbb{Q}(q(g), \sqrt{p})$,*

$$\#X_{G,q}(\overline{K}) \leq 6 \cdot 2^{\lfloor (p-1)/3 \rfloor}, \quad [K(E(g) : g \in G) : K] \leq 6 \cdot 2^{\lfloor (p-1)/3 \rfloor}.$$

Proof. There are at most two choices for e and three for λ . Here $G[3] = \{0\}$, so $m_\lambda - 1 \leq \lfloor (p-1)/3 \rfloor$. There are no Gaussian exceptions for $p \geq 5$. $\square$

These are deliberately coarse explicit upper bounds, not predictions of the actual class-field degrees. Sector-specific dimensions are usually more useful.

5 Complete small-order certificates

5.1 The cyclic group of order three

Use $G = \mathbb{Z}/3\mathbb{Z}$, $s = \sqrt{3}$, $r = (-1 + is)/2$, $q(1) = q(2) = r$, and $K = \mathbb{Q}(s, i) = \mathbb{Q}(\zeta_{12})$. Write $E = (a, b, d)$ and $C = c$. Eliminating the ideal $(P_{xy}, tc - 1)$ with lexicographic order t, c, d, a, b gives

$$K[X_{G,q}] \simeq K[b]/(p(b)),$$

where

$$\begin{aligned} p(b) &= b^5 + 2ib^4 - 2b^3 + (s - i)b^2 + (1 + si)b - (s - i)/2 \\ &= \left(b + \frac{s + i}{2}\right) Q(b), \\ Q(b) &= b^4 + \frac{-s + 3i}{2}b^3 - \frac{1 + si}{2}b^2 + sb + r. \end{aligned}$$

The other coordinates are recovered by

$$\begin{aligned} a &= -ib^4 + \frac{5 + si}{2}b^3 - (s - 2i)b^2 - sib + \frac{3s + i}{2}, \\ d &= ib^4 - (1 + si)b^3 + (s + i)b^2 - (1 - si)b - 2i, \\ c &= -(1 - si)b^4 - \frac{3s + 5i}{2}b^3 + (2 - si)b^2 - (s - 3i)b - \frac{5 + si}{2}, \\ t &= sib^4 - (2s + 2i)b^3 + (3 - si)b^2 + 3ib - \frac{3 + si}{2}. \end{aligned}$$

These polynomials, together with $p(b)$, form the computed reduced Groebner basis in solved form. The script regenerates it directly from the original cubic equations, rather than accepting these formulas as input.

Exact discriminant calculations give

$$\text{disc } Q = -147, \quad Q\left(-\frac{s+i}{2}\right) = r, \quad \text{disc } p = -147r^2 \neq 0.$$

Thus there are exactly five reduced geometric points. The linear factor gives (5). The four roots of Q give the normalized solutions, with $\lambda = (s-i)/2$; substitution modulo Q verifies all of (2)–(4).

One can also see that no normalized solution has $b = d$. In the normalized part the elimination equations give

$$b + d = \frac{a(1-si)}{2}, \quad (1+si)b^2 - 2ab + 1 - si = 0, \quad a^2 - sa - 1 = 0.$$

The discriminant of the quadratic in b is $4(a^2 - 4)$, which is nonzero under the equation for a . Hence only the exceptional point has a constant punctured value, proving the needed order-three part of Proposition 2.2.

The coefficients of p are algebraic integers in K . Consequently $\mathcal{O}_K[1/21][b]/(p)$ is an explicit finite étale model of this generic solution algebra. This is a concrete good-reduction statement outside primes above 3 and 7 for this model, not a general conductor theorem for Stark values. The primitive polynomial gives an algebraic model; identifying its points with analytically selected special values is a separate task.

5.2 The anisotropic four-point metric

For $G = (\mathbb{Z}/2\mathbb{Z})^2$, $q = (1, -1, -1, -1)$, $E = (a, b, d, e)$, the reduced lexicographic Groebner basis of $(P_{xy}, tc - 1)$ in the order t, c, e, d, b, a is

$$t + 1, \quad c + 1, \quad e + 1, \quad d + 1, \quad b + 1, \quad a - 1.$$

Thus this entire metric-specific solution scheme is the single reduced rational point already displayed. The existence theorem in [1] is an existence statement for each underlying two-generator group, not for every quadratic function on that group. There is no contradiction with that theorem.

6 An elementary recovery of a known higher-rank obstruction

Schopieray's earlier classification [6, Theorem 1.1] implies that the Izumi near-group ring for $(\mathbb{Z}/2\mathbb{Z})^r$ with multiplicity 2^r is not categorifiable for $r \geq 3$. Thus the nonexistence conclusion below is *not new as a categorical consequence*. It is useful to recover it straight from the finite equations, without categorical rigidity or a unitarity assumption.

Proposition 6.1. *For every $r \geq 3$ and every metric quadratic function on $G = (\mathbb{Z}/2\mathbb{Z})^r$, equation (1) has no solution with $C \neq 0$.*

Proof. Put $N = 2^r$. Since $N \geq 8$, the corrected normalization has no exception. Every $u \neq 0$ is its own negative, so

$$q(u)^4 = 1, \quad E(u)^2 = q(u)^{-1}, \quad E(u) \in \mu_8.$$

Let $L = \mathbb{Q}(\zeta_8)$. The value $s = \sqrt{N}$ lies in L , and the Fourier matrix has entries $\pm 1/s$. Since qE is nonconstant off zero, subtracting the Fourier equations at two nonzero points with distinct values of qE shows that $\lambda \in L$: the terms involving $E(0)$ cancel and all other terms lie in L . Either Fourier equation then gives $e = E(0) \in L$. Because the two possible values of e are real, $e \in \mathbb{Q}(\sqrt{2})$.

The identity $2e - s = \pm\sqrt{N+4}$ implies $\sqrt{2^{r-2}+1} \in \mathbb{Q}(\sqrt{2})$. The positive integer $2^{r-2}+1$ is odd. An odd integer whose square root belongs to $\mathbb{Q}(\sqrt{2})$ must be a square in $\mathbb{Z}$: its square class cannot be 2. Thus

$$2^{r-2} + 1 = m^2.$$

The consecutive even integers $m-1, m+1$ have product a power of two. They must be 2, 4, so $m=3$, $r=5$, and $N=32$.

It remains to exclude that order. Write $z = \zeta_8$ and $\sqrt{2} = z - z^3$. The Gaussian sum $\sum q(u)^{-1}$ is a Gaussian integer of norm 32. It is therefore one of $\pm 4 \pm 4i$, so Γ is an odd power of z . As $\lambda^3 = \Gamma$ and $\lambda \in L$, it follows that λ is an odd power of z as well. Here the only roots of unity in L are μ_8 ; equivalently, a nontrivial cube root of unity would generate the quadratic subfield $\mathbb{Q}(\sqrt{-3})$, which L does not have.

The Fourier equation at zero now requires a sum of 31 eighth roots of unity to equal

$$\sum_{u \neq 0} E(u) = (4\sqrt{2}\lambda - 1)(2\sqrt{2} \pm 3).$$

Use the rational basis $1, z, z^2, z^3$ of L . Any sum of 31 elements of $\mu_8 = \{\pm 1, \pm z, \pm z^2, \pm z^3\}$ has coefficient ℓ^1 norm at most 31. For $\lambda = z, z^3, z^5, z^7$ the displayed right side has coefficient norm, respectively,

$$37, \quad 47, \quad 47, \quad 37,$$

for either sign. For example when $\lambda = z$ and the sign is positive it is $9 + 14z + 12z^2 + 2z^3$. The other seven expansions are included in the exact verification output. This contradiction excludes $N=32$ as well. $\square$

7 Arithmetic scope

Theorem 3.1 gives a finite étale K -algebra, hence a product of finite separable extensions. The two metrics in Section 5 additionally have explicit primitive presentations. A generic linear combination of the coordinate functions is a primitive element, yielding a squarefree univariate presentation and polynomial recovery of every coordinate. Clearing denominators in this presentation and its inverse, and inverting its discriminant, gives an effectively computable integer $D \neq 0$ and a finite étale model over $\mathcal{O}_K[1/D]$. This provides good reduction and unique infinitesimal lifting outside a finite computable set; no small uniform formula for D is claimed.

These conclusions do not prove algebraic integrality or a unit property, identify the predicted ray class fields, determine the Artin action, or produce unitary Galois conjugates. Algebraicity and the equations already gave a Galois permutation action. Finite étaleness adds infinitesimal and reduction control, not an automatic reciprocity law. These distinctions agree with the open questions in [1, §6].

For an EIT-style program the contribution is methodological: study the actual relation scheme and its tangent spaces, rather than only numerical relations at one point. No polynomial EIT theorem is applied here to the quantum dilogarithm; its categorical relation system is analysed on its own terms.

Reproducibility and research disclosure

The script `scripts/verify_rw_extensions.py` regenerates the exceptional solutions, full-rank tangent minors, and saturated low-order Gröbner bases from the raw equations. It also checks

the finite root-of-unity calculation for $N = 32$. Its output is recorded in the repository. These exact checks prove the stated finite certificates, not the arbitrary-group categorical argument. The separate script `scripts/verify_rigidity_identities.py` checks the corrected scalar coefficient, the unit-rescaling calculation, representative character projectors, and the formal Leavitt contraction in reconstruction. These are exact regression checks with a stated finite scope, not a formalization of the universal realization or monoidal coherence proof. AI-assisted tools were used for drafting, proof exploration, literature comparison, and internal checks. The manuscript has not been independently refereed or formally verified.

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